

Smart-Prep, Explained From Zero

A plain-language lecture on the AI we built — for every teammate, no background needed

The goal of this document. By the end, you can explain our AI to your *lola* — and to a judge. We assume you know **nothing** about business math or AI. Every hard word is explained with an everyday example *before* we name it. Read it slowly, once. Then read §9 (the 5-sentence version) and §11 (your rehearsal lines) until you can say them without looking.

First, the only scary word, removed: a "**model**" is just a **recipe** — a fixed set of steps a computer follows to turn information into an answer. Not a robot. Not magic. A recipe. We use a few small recipes that work together. That's the whole "AI."

A note on the two names: the whole predictive engine is called **SukiSense**. **Smart-Prep** is its *first part* — the spoilage one — and it's the part this lecture explains. So when you hear "SukiSense," think *the engine*; when you hear "Smart-Prep," think *the first thing it does*.

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§1 · The one job (everything starts here)

Every single morning, Tina has to answer one question for her cooked food:

"**How many should I make today?**" — how many siomai, cups of coffee, rice meals, pandesal, lumpia.

It sounds small. It is actually where the most money is won or lost, because:

- Cooked food (ready-to-eat) earns the **best profit** in the store — **45% to 55%**.
- But it **spoils fast**. If she makes too much, the leftover is thrown away — **money in the trash**.
- If she makes too little, she **sells out** and turns customers away — **money she never earned**.

So she's trapped between two mistakes every day:

Mistake	What happens	What it costs
Cook too much	leftovers spoil	she loses the cost of making them
Cook too little	sells out early	she loses the profit she would have made

Our AI, Smart-Prep, has exactly one job: help Tina pick the number that loses the least money — for each item, every day. That's it. If you remember nothing else, remember this paragraph.

§2 · The big picture (the whole AI in 4 simple steps)

Smart-Prep answers the morning question in four steps. The rest of this lecture is just these four steps, slowed down:

1. **GUESS** how many will sell today. (*called "forecasting"*)
2. **ADMIT the guess isn't exact** — figure out the realistic range (could be a little more, could be a little less). (*called "uncertainty"*)
3. **DECIDE the smart number to cook** — using which mistake is more expensive for that item. (*called the "newsvendor" decision*)
4. **LEARN from what really happened** tonight, so tomorrow's guess is better. (*the "learning loop"*)

Picture an experienced *tindera* in her head: "*Fridays are busy, it's payday, looks like rain... I'll make a bit extra coffee but go easy on the pandesal.*" **She is already doing all four steps in her head.** Smart-Prep just does the same thinking with memory and math — consistently, for every item, without forgetting or guessing wrong on a tired morning.

Say this on stage: "*Our AI doesn't replace Tina's instinct — it's her instinct, written down as math, so it never has a bad morning and never forgets a pattern.*"

§3 · STEP 1 — Guessing how many will sell (forecasting)

The problem: before the day starts, we need a best guess of today's demand for each item.

The everyday version: how would you guess today's bread sales? You'd think about three things:

1. **What sells on a normal day** — "lately, about 40 pandesal a day."
2. **Whether business is slowly growing or shrinking** — "this month feels a bit busier than last."
3. **The weekly rhythm** — "Saturdays are always biggest, Sundays quietest."

That's exactly what our main forecasting recipe tracks. Its name is **Holt-Winters** (just two inventors' names — don't be scared of it).

3.1 What Holt-Winters actually does

It keeps a running estimate of those same three things and combines them:

- **The usual level** — roughly how much sells on an ordinary day. It updates this constantly: recent days matter more, old days matter less (but aren't forgotten).
- **The trend** — is the usual level slowly drifting up or down over weeks?
- **The weekly rhythm** — the repeating shape across Monday → Sunday. (*The fancy word for "a pattern that repeats on a fixed cycle" is seasonality. A week is our cycle.*)

One more term, made easy — "**exponential smoothing**." It just means: "**trust recent days more than old days, but don't throw the old days away**." Last week tells you more about tomorrow than three months ago — but the old months still shaped the overall pattern. That's all "smoothing" means: gently blending the past, weighted toward what's recent.

So when someone says our model is "**triple exponential smoothing**," they're saying the *same thing* as Holt-Winters — "triple" because it smooths **three** things (level, trend, weekly rhythm). **Same recipe, two names**. If a judge uses either name, you nod — they mean the same model.

3.2 The payday nudge

In the Philippines, the **15th and 30th** (payday) change everything — people have cash. So we added one nudge: around payday, **push the forecast up**. And the model **learns how big** that bump usually is from past paydays, instead of us guessing. We call it "**payday-aware Holt-Winters**" — Holt-Winters, plus it watches the calendar for payday.

3.3 The backup recipe for week one (the "seasonal baseline")

Holt-Winters needs a few weeks of history to learn the rhythm. **On day one, there is no history**.

Everyday version: a brand-new helper on their first week doesn't know the store's rhythm yet. So they use the simplest sensible rule: "*To guess this Friday, look at the last few Fridays and copy that, then adjust for payday.*" Plain, but it works from day one.

That simple rule is our **seasonal baseline** — the fallback. Once 3–4 weeks of data exist, the smarter Holt-Winters takes over. **Smart-Prep automatically uses whichever one has been more accurate lately** — you don't choose.

Say it simply: *"To guess demand, we use Holt-Winters — it learns the normal level, the slow trend, the weekly rhythm, and the payday bump. Before there's enough data, it falls back to a simple 'same-weekday average.'"*

§4 · STEP 2 — Knowing how wrong the guess could be (uncertainty)

Here is the idea most people miss, and it's simple:

A forecast of "30 coffees" is a **best guess** — **not a promise**. Some days 26 sell. Some days 35. We must know the **realistic range**, not just the middle number — because we cook *before* we know.

Everyday version: the weather app doesn't say "it will rain." It says "**30% chance of rain**." It gives you the *odds and the range* so you can decide whether to bring an umbrella. Same here: Smart-Prep says "*probably about 30 coffees, realistically somewhere from 25 to 36.*" That **range** is what lets us safely cook a little extra to cover the busy-day chance.

4.1 Why we use different "shapes" of range

Different items vary in different ways, so the *shape* of their range differs. We pick the shape that fits each item. There are three, and you only need the intuition:

Shape (the name)	Use it for	Everyday picture
Normal ("bell curve")	big, steady sellers — coffee (~28/day), pandesal (~40/day)	like people's heights: most cluster near the average, a few higher/lower, evenly balanced both sides
Poisson	small-count items — rice meals (~8/day)	like counting how many tricycles pass in 10 minutes : small whole numbers, can't go below zero, naturally a bit lopsided
Negative Binomial	small-count items that are extra jumpy	a usually-quiet item that sometimes gets a surprise group order — bigger swings than Poisson expects

Why this matters in one line: you can't measure a big steady seller (coffee) and a tiny jumpy one (rice meals) with the same ruler. Smart-Prep picks the right ruler per item, **automatically**, from each item's own history.

You do NOT need to do any of this math. The computer chooses the shape. You only need to be able to say *why*: **big steady sellers** → bell curve; **small counts** → Poisson; **jumpy small counts** → **Negative Binomial**. That sentence alone answers the judge.

Say it simply: "We don't just predict an average — we estimate the realistic range of demand, and we use the right kind of range for each item depending on whether it sells in big steady numbers or small jumpy ones."

§5 · STEP 3 — Deciding the smart number to cook (the newsvendor)

This is the **heart** of the AI and the part judges love. It's built on one insight:

The two mistakes do NOT cost the same. So the smart amount to cook depends on *which mistake is more expensive for that specific item.*

The classic everyday version (this is literally where the name comes from): a man sells **newspapers** on the corner. Each morning he buys his stack. Papers unsold by night = wasted money. But if he runs out by noon, he misses sales he could've had. **How many should he buy?** It depends on his profit per paper versus his loss per unsold paper. That puzzle is called the "**newsvendor problem**," and it's the standard, textbook method for anything that spoils — bread, coffee, newspapers, flowers. **We're using a 70-year-old, proven business formula — not a buzzword.**

5.1 Watch it decide, with Tina's real numbers

The rule: **if missing a sale hurts more than wasting one → cook a bit MORE than the average. If wasting hurts more than missing → cook a bit LESS.**

Coffee — sells ₱30, costs ₱12.

- Miss a sale → lose the **profit, ₱18.**
- Waste a cup → lose the **cost, ₱12.**
- Missing (₱18) **hurts more** than wasting (₱12). → **Cook a little MORE than average.** (If the guess is 32, cook ~33 — "cook 33, not 34.")

Pandesal — sells ₱4, costs ₱2.40.

- Miss a sale → lose the **profit, ₱1.60.**
- Waste one → lose the **cost, ₱2.40.**
- Wasting (₱2.40) **hurts more** than missing (₱1.60). → **Cook a little LESS than average.**

Same store, opposite decisions — and both are correct. That's the magic sentence: **Smart-Prep makes a different, money-based decision for every item.** It is the opposite of "cook 20% extra of everything."

5.2 The one number behind it (optional, for the curious)

Smart-Prep turns "which mistake costs more" into a score between 0 and 1 called the **critical ratio** — *the share of the total mistake-cost that comes from missing a sale.*

critical ratio = $\text{profit-if-sold} \div (\text{profit-if-sold} + \text{cost-if-spoiled})$

- Coffee: $18 \div (18 + 12) = \mathbf{0.60}$ → high → aim **above** average (cook generously).
- Pandesal: $1.60 \div (1.60 + 2.40) = \mathbf{0.40}$ → low → aim **below** average (cook lean).

Plain meaning: a higher score = "this item is worth protecting against sell-outs, so cook more." A lower score = "this item is cheap to miss but costly to waste, so cook less." You don't need the formula on stage — just "*high-margin items we cook generously, cheap-to-waste items we cook lean, and the model calculates exactly how much.*"

Say it simply: *"The newsvendor model picks the prep quantity that loses the least money — balancing the cost of waste against the cost of a missed sale, separately for each item."*

§6 · STEP 4 — It gets smarter every day (the learning loop)

The AI is not "set once." It improves nightly. Here's the loop:

- **Morning:** Smart-Prep recommends — *"make 33 coffee, 25 siomai, 8 rice meals..."*
- **During the day:** customers buy.
- **At close (30 seconds):** Tina records four things — **made, sold, spoiled, and did it sell out? (and what time).** (*This little daily record is the single most important habit — without it, nothing can learn.*)
- **Overnight:** the AI compares its guess to what really happened and adjusts:
- Made too much, lots spoiled → **aim lower** tomorrow.
- Sold out early → **aim higher** tomorrow.
- Payday clearly spiked sales → **remember that** for next payday.

So it learns from its own mistakes, a little every day. **That's the honest meaning of "the AI learns" — not a brain, just a recipe that corrects itself with new data.**

6.1 The clever bit: the "sold-out trap" (censoring)

Here's a subtle problem that separates a real system from a toy:

If Tina made **30** coffees and they **sold out by 2pm**, how many did people actually *want*? We don't know — maybe 30, maybe 45. **We only saw the ones who got served.**

Everyday version: a jeepney passes your stop already **full**. How many *more* people wanted to ride? You can't see them — they're left at the curb. If you only count who fit, you'll always **underestimate** demand.

If the AI naively learns "sold 30," it thinks demand was 30, so it suggests ~30 again, sells out again, and **spirals downward forever**. Smart-Prep avoids this: when an item **sells out**, it flags it and **estimates the demand it didn't get to see** (using the sell-out time). In our testing, ignoring this made demand look **17% too low**; correcting it brought the estimate **within about 1%** of the truth.

Say it simply: *"When an item sells out, the real demand was higher than what we sold — like a full jeepney passing a crowded stop. We correct for that, so the forecast doesn't shrink itself into a corner."*

§7 · The fancy AI we tested — and chose NOT to use (this is a strength)

You will be asked: *"Why not use real machine learning / deep learning / AI like ChatGPT?"* Here's the honest, confident answer.

We **did** test a modern machine-learning model — it's called **LightGBM** (a popular, powerful "learns-from-many-clues" model). The catch: powerful models like that need **lots and lots** of data to shine — years of it, ideally many stores.

Tina has about **four months** of data from **one** store. With that little, LightGBM actually did **worse** than our simple Holt-Winters. So we kept the simpler recipe — because here it is **both more accurate AND explainable** (we can tell Tina exactly *why* it recommended a number; a big ML model is a "black box" that can't).

Say it on stage (this scores points): *"We tested a modern machine-learning model, LightGBM. With Tina's small dataset it performed worse than the simpler method, so we chose the simpler, explainable one. We used the right-sized tool — and we proved it by testing, instead of using fancy AI just to sound impressive."*

This is exactly the maturity the case asks for: *"Buzzwords score nothing."*

§8 · How good is it? (the results, in plain words)

We **backtested** it — meaning we replayed history and checked the AI's guesses against what *actually* sold (a fair, honest exam). The headline results:

Plain question	Answer
How accurate is the forecast?	On average it's off by about 19% (<i>lower is better; the simple model beat LightGBM's 20%</i>)
How much less food is wasted?	Spoilage drops from about 16% → 9% of what's prepped
How much money does that protect?	about ₱24,000/year now, ~₱47,000/year at full SukiMart size
What does it cost to run?	₱0 extra — it's a feature of the POS she's buying anyway

"**WAPE**" is just the accuracy score — "*on average, how far off were our guesses, as a percentage.*" 19% off. Lower is better. You can say "our average forecast error" instead of the acronym.

§9 · The whole thing in 5 sentences (memorize this)

1. Every morning Tina must decide how much cooked food to make — too much spoils, too little sells out.
 2. **Smart-Prep** guesses **tomorrow's demand** for each item using **Holt-Winters**, which learns the normal level, the weekly rhythm, and the payday bump.
 3. It then estimates a **realistic range** around that guess (a bell curve for big sellers, a counting model for small-count items).
 4. It picks the **profit-smart amount to cook** with the **newsvendor model** — cook *more* of high-margin items like coffee, *less* of cheap-to-waste items like pandesal.
 5. Every night it **learns** from what sold and spoiled (and corrects for sell-outs), getting better over time — and we proved fancier AI wasn't worth it here.
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§10 · Glossary — every scary word in one plain line

Word	What it really means
Model	a recipe: fixed steps that turn data into an answer
Forecast	a best guess of how many will sell
Holt-Winters	the forecasting recipe that learns level + trend + weekly rhythm
Triple exponential smoothing	the <i>exact same thing</i> as Holt-Winters (three things smoothed)
Exponential smoothing	"trust recent days more than old days, but don't forget the old ones"
Seasonality	a pattern that repeats on a cycle (for us, the weekly Mon → Sun shape)
Seasonal baseline	the simple backup forecast for week one ("average the last few same-weekdays")
Uncertainty / distribution	the realistic <i>range</i> of demand, not just the average
Normal (bell curve)	the range-shape for big, steady sellers
Poisson	the range-shape for small whole-number counts (e.g., 8 rice meals)
Negative Binomial	like Poisson but for small counts that swing more than expected
Newsvendor model	the decision recipe: cook the amount that loses the least money
Critical ratio	a 0–1 score for "how generously to cook this item" (high = cook more)
Censoring / the sold-out trap	sell-outs hide true demand (the full-jeepney problem); we correct for it
Backtest	replaying history to fairly test how good the guesses were
WAPE	the accuracy score: average forecast error as a % (lower is better)
LightGBM	a powerful machine-learning model we tested but didn't use (needs more data)

§11 · Your turn — rehearsal lines (say these out loud)

If a teammate or judge asks "what is the AI?":

"It's called Smart-Prep. Every morning it tells Tina how many of each ready-to-eat item to cook — enough to not lose customers, but not so much that food gets wasted. It guesses demand, estimates the range, then picks the money-smart amount to cook."

If asked "is it really AI, or just an average?":

"Not an average. It learns weekly and payday patterns, estimates the realistic range per item, and makes a different cost-based decision for each one — more coffee because a missed sale costs more, less pandesal because waste costs more. And it improves every day from what actually sold."

If asked "what models?":

"Holt-Winters to forecast demand, a Normal or Poisson range to handle uncertainty, and the newsvendor model to choose the profit-best quantity. We tested LightGBM too, but it didn't beat the simpler model on Tina's data."

The 3 things to NEVER forget:

1. **One job:** how much to cook — waste vs. sell-out.
2. **Three steps:** guess (Holt-Winters) → range (Normal/Poisson) → decide (newsvendor).
3. **Per-item, money-based:** cook *more* coffee, *less* pandesal — and it learns nightly.

You don't need to be a data scientist. You need to explain a smart morning decision in plain Filipino. If you can tell the jeepney story and the coffee-vs-pandesal story, you understand our AI better than most people in the room.